

THE FIREWALL GUIDE

FOR NETWORKING AND CYBERSECURITY STUDENTS

The firewall you already have at home: your fiber-optic router

Real-world case study: Fios Router (Verizon)

Level: Junior – Intermediate

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1. What is a firewall?

A firewall is a security system — hardware, software, or both — designed to control and monitor incoming and outgoing network traffic. Its main function is to act as a protective barrier between a secure internal network (your home, your lab, an office) and untrusted external networks, such as the Internet. It blocks unauthorized access, malware, and cyberattacks, while allowing legitimate traffic to pass through.

A firewall works based on a set of predefined security rules that determine what type of traffic is allowed and what gets rejected. It isn't a magic "black box": it's decision logic applied packet by packet, or connection by connection, depending on the type of firewall.

1.1 Types of firewalls

Type	How it works
Packet filtering	Examines individual data packets and blocks or allows them based on IP addresses and ports.
Stateful inspection	Tracks the state of active connections to make smarter decisions about what traffic to allow. This is the type your home router uses.
Application firewall (Proxy)	Acts as an intermediary between users and services, inspecting traffic at the application level.
Next-Generation Firewall (NGFW)	Combines traditional functions with advanced capabilities: deep packet inspection (DPI), intrusion prevention (IPS), and application-based filtering.

🔗 Why start here

You don't need an expensive lab to learn real firewalls: the router your fiber-optic provider gave you already has one built in (a stateful type). Understanding its options is the real first step toward pfSense, Cisco ASA, or any enterprise NGFW: the logic — inbound vs. outbound traffic, ports, NAT — is the same.

2. The study device: a fiber-optic router

For this guide we use a Verizon Fios Router as our example — the router the fiber-optic company itself provides bundled with the service (it combines an ONT modem + router + firewall in a single device). This kind of router is representative of most current ISP equipment (other providers included): they all expose a web panel with very similar sections: General, Access Control, Port Forwarding, Port Triggering, DMZ Host, Remote Administration, Static NAT, and Security Log.

2.1 How to access the firewall panel

- Connect to the router's network by cable or WiFi (not from outside your home).

- Open a browser and go to the default gateway address, usually 192.168.1.1 (on Fios routers) — on other brands it may be 192.168.0.1 or 10.0.0.1.
- Enter the administrator credentials. They're usually printed on a physical label on the router itself (default username and password).
- Once inside, look for the Firewall tab in the top menu. That's where every option covered in this guide lives.

🔒 First security finding

If you've never changed the router's default administrator password, anyone who connects to your WiFi (or guesses it) can get into the full panel and reconfigure your firewall. Changing that password to a strong, unique one is the highest-payoff hardening task there is, and it costs nothing.

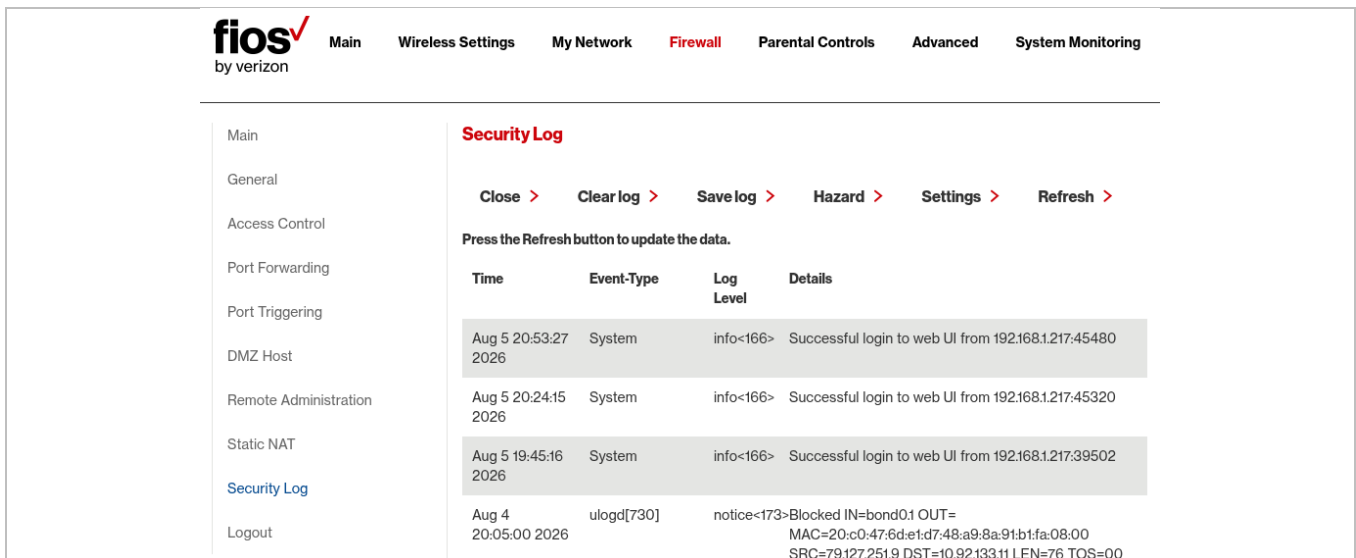


Fig. 1 — Fios Router admin panel. The Firewall menu (in red) groups all the sections we'll cover.

3. General security levels (Inbound / Outbound)

Before touching any advanced option, every router firewall lets you choose a general security posture. It's defined with two key concepts you'll keep running into on any professional firewall you use later:

- Inbound Policy: connections that try to enter your network from the Internet — for example, someone trying to reach your computer from outside.
- Outbound Policy: connections that your own devices initiate toward the Internet — opening a browser, downloading updates, etc.

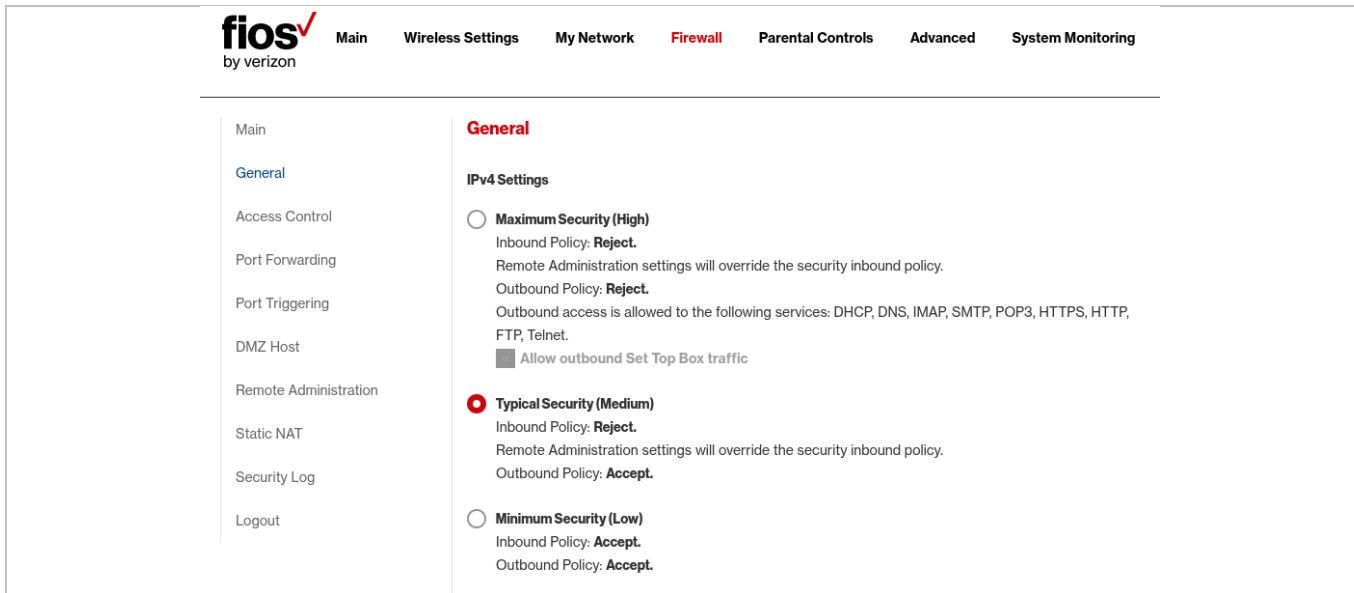


Fig. 2 — Security level selector. "Typical Security" is the active option on this router.

Level	Inbound / Outbound	When to use it
Maximum Security (High)	Reject / Reject (with exceptions: HTTP/HTTPS, DNS, email)	A very strict environment. Gives strong security, but tends to break applications that use non-standard ports (online games, VPNs, specific security tools, P2P clients).
Typical Security (Medium)	Reject / Accept	The recommended setting almost always: blocks all unsolicited traffic from the Internet, without restricting what you initiate yourself. Real security with no daily friction.
Minimum Security (Low)	Accept / Accept	The firewall is essentially off: everything goes in and out unfiltered. Never recommended, unless you have another dedicated firewall behind it handling real security.

What an attacker sees at each level

With "Minimum Security" a port scan from the Internet (nmap, masscan) sees your entire network as if there were no firewall at all: any misconfigured service on your devices is directly exposed. With "Typical", that same scan finds nothing responding from outside, because inbound traffic is set to Reject. That's literally the firewall's job: making the reconnaissance phase of an attack fail from the start.

4. Access Control

Lets you block Internet access for specific devices within your local network (LAN), identifying them by IP address or MAC address, and choosing which protocols to block.

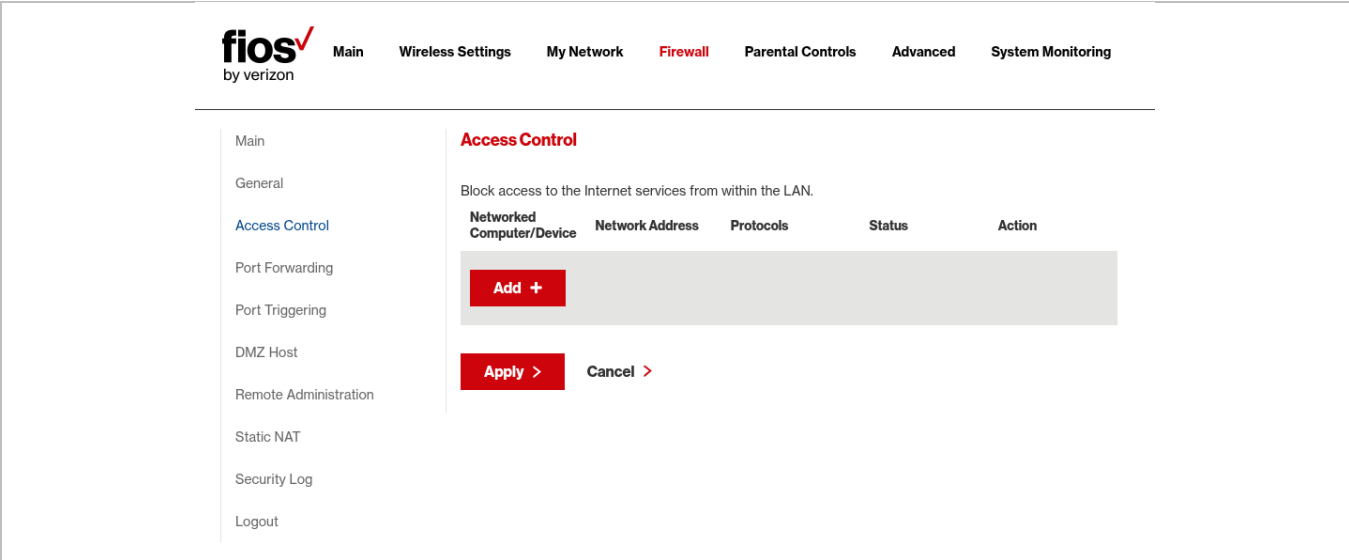


Fig. 3 — Access Control screen: no rules created yet.

Why use it?

It's very useful if you have a device running test software or a vulnerable device (an old IoT gadget, a pentesting VM) that you need to isolate so it can't communicate with the outside world unsupervised. It's a cheap form of microsegmentation for a home network that doesn't have VLANs.

5. Port Forwarding

Opens a specific "door" in your firewall so traffic coming from the Internet can pass through it and reach a particular device on your network. You're telling the router: "any traffic arriving on port 22, send it exclusively to my laptop's local IP."

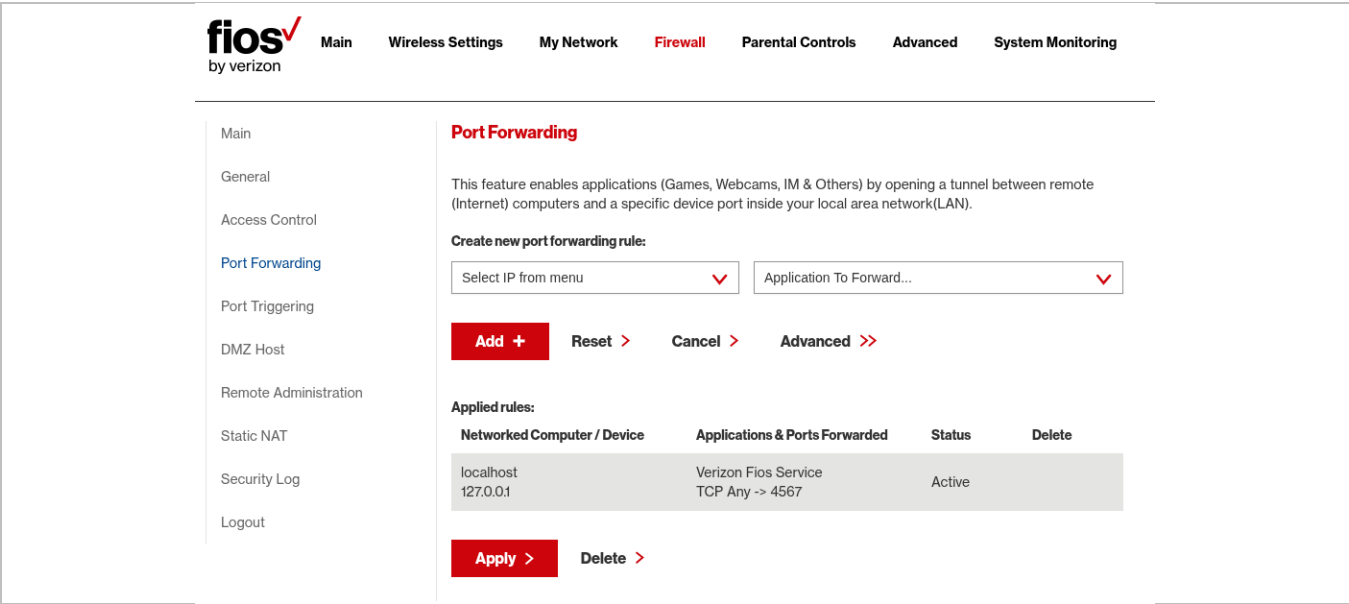


Fig. 4 — Port Forwarding: here's an active rule (Verizon Fios Service, TCP port 4567) used by the provider itself.

Why use it?

It's essential if you decide to host your own projects, set up a practice lab, or need to connect via SSH to a Linux environment from outside your home.

How this helps an attacker

Every port opened with Port Forwarding is a permanent door into an internal device, 24 hours a day. If you open port 22 (SSH) without changing the default password or without using keys, you become a target for bots that scan the entire Internet looking for exactly that open port to try brute-force attacks. Golden rule: open only what's strictly necessary, and close the rule once you're done with the lab.

6. Port Triggering (Dynamic Port Forwarding)

A "smarter", temporary, and safer version of classic Port Forwarding: a rule that tells the router to open certain inbound ports only when a device on your local network starts an outbound connection on a specific port (the "trigger" port).

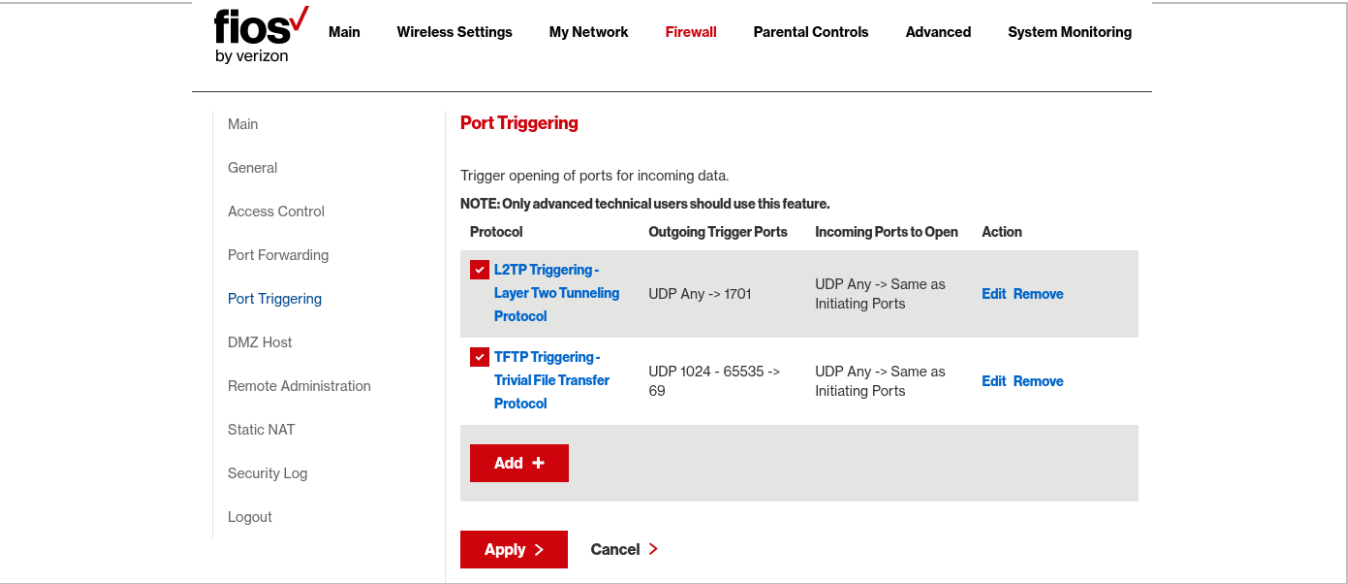


Fig. 5 — Port Triggering rules for L2TP and TFTP.

How does it work?

Imagine you're playing an online game. Your device sends data to an external server over port 5000 (the trigger). The router detects it and says: "someone inside is using this application, I'll temporarily open inbound port 6000 to let the response through."

	Port Forwarding	Port Triggering
Exposure window	The door stays open 24/7	Only opens while the application is in use
Risk level	Higher: more time to be scanned	Lower: closes itself when done

It's ideal for video games, video conferencing, or messaging (SIP or VoIP services), especially when several devices at home might need to use the same application at different times: the router assigns the rule dynamically to whichever device requests it first.

7. DMZ Host (Demilitarized Zone)

Exposes a device on your internal network completely to the Internet, bypassing all of the firewall's protection rules.

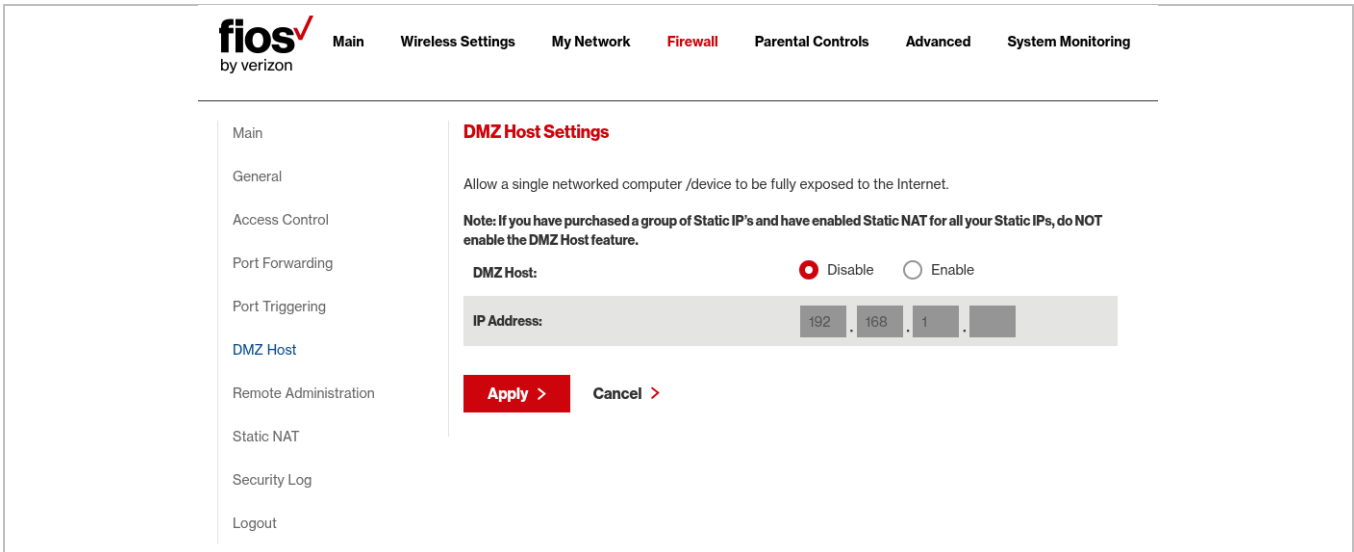


Fig. 6 — DMZ Host disabled, the recommended default configuration.

Recommendation

Avoid using it. Placing a device in the DMZ turns it into an immediate target for automated scans and external attacks, because it literally removes the protective barrier for that device. It's only used in very controlled environments to isolate critical services, and almost never in a home or study network.

Difference from Port Forwarding and Static NAT

- Port Forwarding: opens specific "doors" (ports).
- DMZ: exposes an entire device to your home's main public IP (all ports, unfiltered).
- Static NAT: links a dedicated public IP to an internal device — requires the ISP to sell you multiple public IPs.

8. Remote Administration

Allows access to the router's configuration panel from anywhere on the Internet, instead of only while connected to the local network.

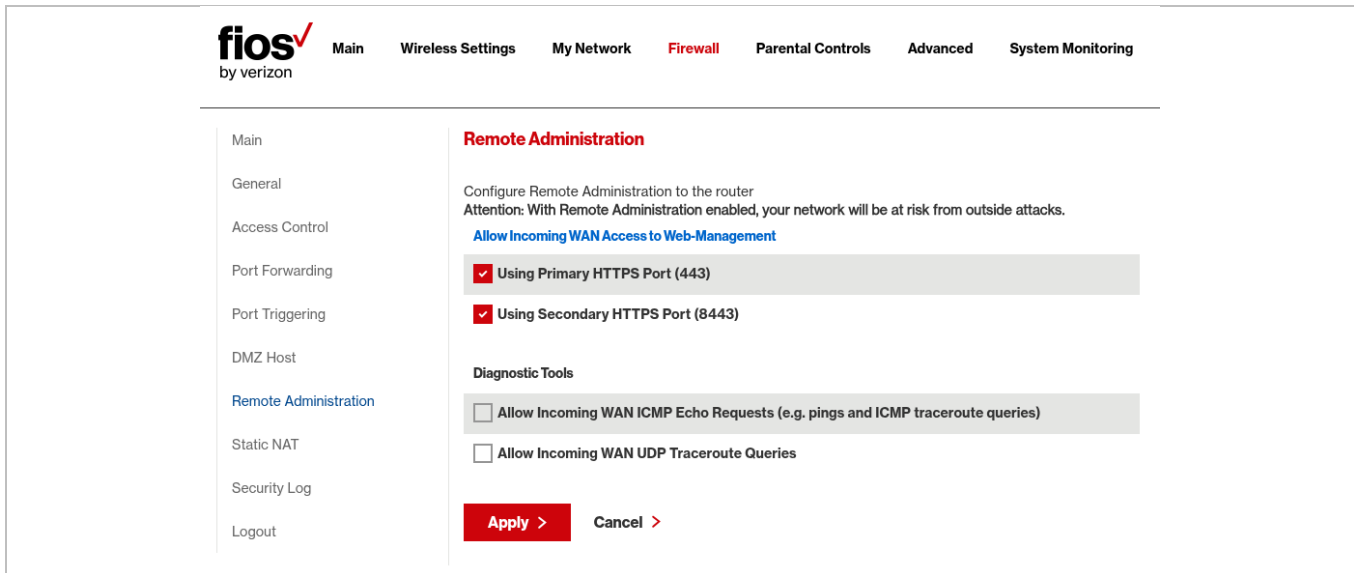


Fig. 7 — Remote Administration: HTTPS ports 443 and 8443 configured for web management access.

🚫 The single most dangerous option in the whole firewall

With Remote Administration on, your router's login page becomes exposed and reachable from anywhere in the world. This lets any automated attacker attempt to brute-force your password without even being near your home. Unchecking these boxes ensures only locally connected devices can access the panel. If you genuinely need to administer the router remotely, the safe alternative is a VPN into your own network — never exposing the panel directly.

8.1 Diagnostic tools: ICMP and traceroute

The same remote-administration section usually holds two checkboxes related to network reconnaissance:

- Allow Incoming WAN ICMP Echo Requests (pings): in the early reconnaissance phase of an attack, a fundamental technique for mapping targets is sending mass "pings" across the Internet to discover which machines are up. Turning this response off makes your network "invisible" to most automated scans, because it simply won't answer to confirm it exists.
- Allow Incoming WAN UDP Traceroute Queries: similar to ping, this lets an outside party trace the exact path packets take to your gateway. Disabling it sharply limits the fingerprint your network exposes outward.

Turning off both checkboxes is a quick, simple, and highly effective hardening measure.

9. Security Log

A detailed history of firewall events: which connections were blocked or allowed, with source IP, ports, and protocol.

Why is it interesting for a cybersecurity student?

Reviewing these logs is an excellent way to understand your network's real-world behavior. Analyzing the external IP addresses that try to interact with your router and get rejected gives you first-hand material to

investigate threats (looking up the IP in OSINT tools, identifying the country or ASN of origin) and practice defensive monitoring concepts (blue team) without leaving home.

10. Static NAT

Also known as 1-to-1 NAT. A much more radical configuration, generally reserved for corporate environments, not home networks.

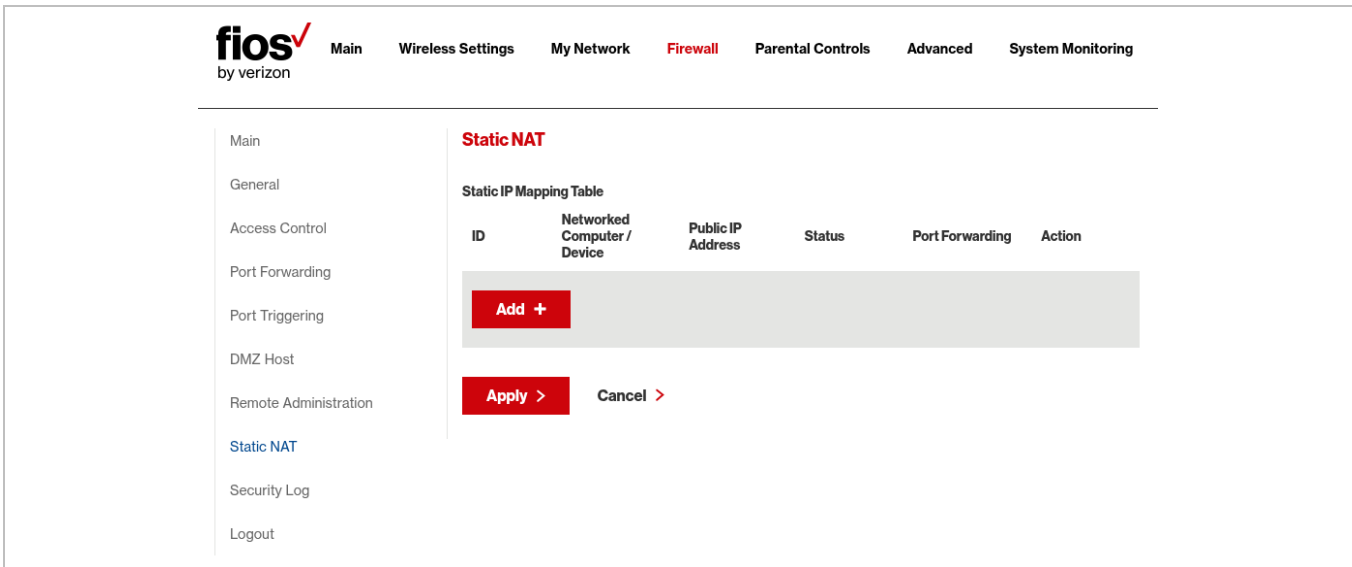


Fig. 8 — Static NAT table, empty by default: this typically doesn't apply at home.

Permanently links an entire public IP address (the one your Internet provider gives you, visible to the world) to a single private IP address of an internal device. All traffic arriving at that public IP is sent directly to that device, with no port filtering: it's as if that device were connected directly to the street, with no protective shield from the router.

Almost never used at home. Businesses use it when they have several physical servers (a web server and a mail server, for example) and need each one to have its own independent public IP address.

11. Summary: which settings make it easier for an attacker to get in

This table ranks the options from riskiest to safest, based on how much easier you're making it for an outside attacker scanning the Internet for easy targets.

Risk	Setting	Why it speeds up an attack
Critical	DMZ Host enabled	Removes the firewall completely for that device: every port is open to the Internet with no filtering at all.
Critical	Minimum Security (Accept/Accept)	The firewall is off: any misconfigured service on your network is visible and reachable from outside.

High	Remote Administration + ICMP/traceroute enabled	Exposes the login panel itself to brute-force attempts from anywhere in the world, and makes initial reconnaissance easier (discovering that your network "exists and responds").
Medium	Port Forwarding without a real need	Every open port is a permanent door; without solid credentials behind it, it becomes a target for bots scanning the Internet for that exact port.
Low	Port Triggering instead of Port Forwarding	The door only opens while the application is in use; the exposure window is minimal.
Minimal	Typical Security + Remote Admin off + ICMP off	Your network doesn't respond to external scans and doesn't expose the admin panel: the attacker's reconnaissance phase fails from the very start.

12. Quick hardening checklist

Apply this to your own router today, whatever the model or ISP:

- ☐ Change the router's default administrator password.
- ☐ Leave the general security level on Typical Security (Reject inbound / Accept outbound).
- ☐ Confirm DMZ Host is set to Disable.
- ☐ Turn off Remote Administration, unless you truly need it (and if you do, use a VPN instead of exposing the panel).
- ☐ Turn off incoming ICMP Echo Requests and UDP Traceroute.
- ☐ Review Port Forwarding and remove rules you no longer use.
- ☐ Prefer Port Triggering over Port Forwarding whenever the application supports it.
- ☐ Check the Security Log every so often to understand what's trying to reach your network from outside.

Next step

Once you've mastered these options on your home router, the natural next step is replicating this same logic (security levels, NAT, port forwarding, DMZ) on a real firewall like pfSense or OPNsense, where you'll have full control and can inspect traffic in much more detail. That's exactly the Project 3 we talked about for your portfolio.